Factor-base designs: D against b on p_1 = 786433 (run-20261015T1657Z)

Kind: solver-axis stage diagnostic (AGENTS.md §2). Observations only: nothing here prices a pipeline, reports S or shows a speedup. The protocol is `research/factor_base_designs_20261015/PREREGISTRATION.md` §1–§3 with Amendment 1 (registered before the first data row).

Run

- **Command:** `target/release/examples/fb_designs_slope run --arms A0,N,A1z,A1g,A4,A11 --m 2,3 --out-dir research/factor_base_designs_20261015/runs` (defaults: t = 2 … 18, 2 planted targets per cell, F4 budget 300 s, hard kill at 420 s, `ulimit -v 13000000` per child process).
- **Commit measured:** `a698ba0399836c56cba11a622737d6c42f418e11` (recorded in every row; clean for `examples/`, `src/` and Cargo files).
- **Host:** "vm", Intel Xeon @ 2.80 GHz, 4 cores, 15 GB RAM, a Claude Code cloud container. **Isolation:** none beyond one system at a time. F4 uses every core, and the instrument checks ran before the study, not alongside it. The times below are wall-clock practicality notes, not the metric.
- **Window:** 2026-10-15, 16:57Z to about 19:42Z (UTC).
- **Instrument:** `f4_fp::f4`, grevlex, degree bound n + d_{max} + 6, D = `solving_degree_max`, with Gebauer–Möller criteria on (the current default; Amendment 1 item 1).
- **Seeds:** cell seed `0x46424453 ⊕ arm≪40 ⊕ m≪32 ⊕ t≪16` draws the set and the curve. Target k is planted from `cell ⊕ (k+1)≪8`.
- **Records:** `runs/<arm>-m<m>.jsonl` and `.log` beside them, `runs/analysis.txt` (native `analyze`), `runs/instrument-check.txt` and `runs/instrument-check-gm0.txt`.

Instrument check (§5 stop rule)

A0 was rebuilt with the pilot's seed rule and draw order (`repro`): 20 planted systems, m = 2 with t ≤ 7 and m = 3 with t ≤ 4. On every system, D, x_R, the curve, the tower, the unknown and equation counts, `inconsistent` and `planted_ok` equal every committed pilot row in `research/pkm_tower_pilot_20260924/runs/`.

`degree_reached` is lower than the pilot's on 16 of the 20 systems (the certification tail). With `F4_FP_GM=0`, all 20 systems match on every field, `degree_reached` included. The stop rule does not fire.

This study's own A0 series (FBDS seeds, so other towers, curves and targets) has the same D as the pilot's planted rows at every (m, t) both have: m = 2, t = 2–7 and m = 3, t = 2–3. At m = 3, t = 4, every pilot row timed out.

Graph

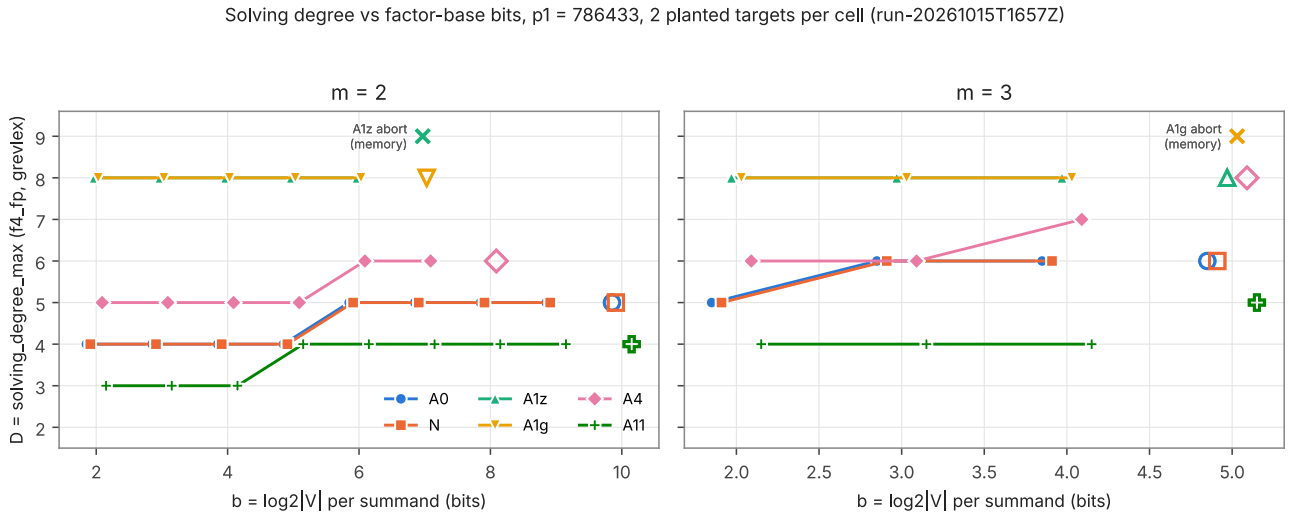


Figure 1: Measured D (`solving_degree_max`) against b = $\log_2|V|$ per summand, per arm, m = 2 (left) and m = 3 (right). Each completed system is plotted, both targets per cell (n = 2 per cell; the two coincide in every cell). Open markers: systems stopped by the 300 s F4 budget, at their partial D (a lower bound). Crosses: memory aborts, no D. Run run-20261015T1657Z, rows from `runs/*.jsonl`, plotted by `plot.py`.

σ and the preregistered verdict

From `runs/analysis.txt`. σ is the least-squares slope of the cell D (the mean of the cell's 2 systems) on b, over completed cells.

arm	m	completed cells (t)	σ	verdict (§1 thresholds)
A0	2	8 (2–9)	0.1905	candidate

A0	3	3 (2–4)	0.5000	no middle ground
N	2	8 (2–9)	0.1905	candidate
N	3	3 (2–4)	0.5000	no middle ground
A1z	2	5 (2–6)	0.0000	candidate
A1z	3	3 (2–4)	0.0000	candidate
A1g	2	5 (2–6)	0.0000	candidate
A1g	3	3 (2–4)	0.0000	candidate
A4	2	6 (2–7)	0.2286	candidate
A4	3	3 (2–4)	0.5000	no middle ground
A11	2	8 (2–9)	0.1786	candidate
A11	3	3 (2–4)	0.0000	candidate

Observations, stated as recorded:

- The slope rule labels the incumbent A0 itself "candidate" at m = 2, where D runs 4, 4, 4, 4, 5, 5, 5, 5 over t = 2–9. At m = 3, A0′s three completed cells read 5, 6, 6.
- N has the same D as A0 at every completed cell, at both arities.
- A1z and A1g have D = 8 at every completed cell. 8 is also the highest input degree of their systems: x is quadratic in the unknowns, so S_3 has degree 8. σ = 0 there means D stays at the input degree over b = 2–6 (m = 2) and b = 2–4 (m = 3). At equal t, A1z − A0 is +2 to +4, outside the predicted ±1. A1g − A1z is 0 at every shared cell, within the predicted ±1.
- A4 had no collisions at any t: $|V| = 2^t$, so b = t. At equal t, A4 − A0 is +1 in every shared cell but one (m = 3, t = 3: 0). Against A0 at $\lceil t/2 \rceil$, A4 is +1 to +2.
- A11′s D is at or below every other arm's at every shared t: 3–4 at m = 2, where it ties A0 and N at t = 5, and 4 at m = 3. Its highest input degree is 2 (m = 2) and 3 (m = 3), with 3 + 3t and 5 + 5t unknowns.
- §5 says an arm reading "candidate" on p_1 is replicated on $p_2 = 2013265921$ before it is described as more than a single-prime reading. No p_2 run was made here. At m = 3, every series has exactly 3 completed cells, the minimum the thresholds accept.

Every system

D, time and status per target (target 0 / target 1). Time is F4 wall seconds, or process wall seconds for an abort. Status is ok, timeout (F4 budget of 300 s) or aborted (memory allocation failed under `ulimit -v 13000000`, SIGABRT). Every series stopped at its first limit cell (Amendment 1 item 5). No planted-solution violation occurred.

arm	m	t	|V|	vars	eqs	D	time (s)	status
A0	2	2	4	4	5	4 / 4	0.0 / 0.0	ok, ok
A0	2	3	8	6	7	4 / 4	0.0 / 0.0	ok, ok
A0	2	4	16	8	9	4 / 4	0.0 / 0.0	ok, ok
A0	2	5	32	10	11	4 / 4	0.1 / 0.1	ok, ok
A0	2	6	64	12	13	5 / 5	0.3 / 0.3	ok, ok
A0	2	7	128	14	15	5 / 5	2.0 / 2.0	ok, ok
A0	2	8	256	16	17	5 / 5	12.9 / 12.9	ok, ok
A0	2	9	512	18	19	5 / 5	92.2 / 92.0	ok, ok
A0	2	10	1024	20	21	5 / 5	302.0 / 302.3	timeout, timeout
A0	3	2	4	7	8	5 / 5	0.0 / 0.0	ok, ok
A0	3	3	8	10	11	6 / 6	0.6 / 0.6	ok, ok
A0	3	4	16	13	14	6 / 6	16.6 / 14.9	ok, ok
A0	3	5	32	16	17	6 / 6	304.2 / 301.6	timeout, timeout
N	2	2	4	4	5	4 / 4	0.0 / 0.0	ok, ok
N	2	3	8	6	7	4 / 4	0.0 / 0.0	ok, ok
N	2	4	16	8	9	4 / 4	0.0 / 0.0	ok, ok
N	2	5	32	10	11	4 / 4	0.1 / 0.0	ok, ok
N	2	6	64	12	13	5 / 5	0.3 / 0.4	ok, ok
N	2	7	128	14	15	5 / 5	2.2 / 2.0	ok, ok
N	2	8	256	16	17	5 / 5	14.8 / 15.0	ok, ok

N	2	9	512	18	19	5 / 5	97.2 / 99.4	ok, ok
N	2	10	1024	20	21	5 / 5	302.5 / 302.6	timeout, timeout
N	3	2	4	7	8	5 / 5	0.0 / 0.0	ok, ok
N	3	3	8	10	11	6 / 6	0.9 / 1.1	ok, ok
N	3	4	16	13	14	6 / 6	24.7 / 24.5	ok, ok
N	3	5	32	16	17	6 / 6	301.3 / 302.2	timeout, timeout
A1z	2	2	4	6	7	8 / 8	0.0 / 0.0	ok, ok
A1z	2	3	8	10	11	8 / 8	0.0 / 0.0	ok, ok
A1z	2	4	16	14	15	8 / 8	0.1 / 0.1	ok, ok
A1z	2	5	32	18	19	8 / 8	2.1 / 2.4	ok, ok
A1z	2	6	64	22	23	8 / 8	110.2 / 106.0	ok, ok
A1z	2	7	128	26	27	– / –	231.2 / 210.9	aborted, aborted
A1z	3	2	4	10	11	8 / 8	0.0 / 0.0	ok, ok
A1z	3	3	8	16	17	8 / 8	1.1 / 1.0	ok, ok
A1z	3	4	16	22	23	8 / 8	237.5 / 247.9	ok, ok
A1z	3	5	32	28	29	8 / 8	320.9 / 322.0	timeout, timeout
A1g	2	2	4	6	7	8 / 8	0.0 / 0.0	ok, ok
A1g	2	3	8	10	11	8 / 8	0.0 / 0.0	ok, ok
A1g	2	4	16	14	15	8 / 8	0.1 / 0.2	ok, ok
A1g	2	5	32	18	19	8 / 8	1.7 / 1.4	ok, ok
A1g	2	6	64	22	23	8 / 8	69.8 / 69.0	ok, ok
A1g	2	7	128	26	27	8 / 8	301.4 / 300.7	timeout, timeout
A1g	3	2	4	10	11	8 / 8	0.0 / 0.0	ok, ok
A1g	3	3	8	16	17	8 / 8	1.2 / 1.1	ok, ok
A1g	3	4	16	22	23	8 / 8	281.3 / 266.4	ok, ok
A1g	3	5	32	28	29	– / –	310.8 / 311.6	aborted, aborted
A4	2	2	4	6	7	5 / 5	0.0 / 0.0	ok, ok
A4	2	3	8	8	9	5 / 5	0.0 / 0.0	ok, ok
A4	2	4	16	10	11	5 / 5	0.0 / 0.0	ok, ok
A4	2	5	32	12	13	5 / 5	0.5 / 0.4	ok, ok
A4	2	6	64	14	15	6 / 6	2.6 / 2.3	ok, ok
A4	2	7	128	16	17	6 / 6	83.7 / 79.8	ok, ok
A4	2	8	256	18	19	6 / 6	301.1 / 301.8	timeout, timeout
A4	3	2	4	10	11	6 / 6	0.0 / 0.0	ok, ok
A4	3	3	8	13	14	6 / 6	0.3 / 0.3	ok, ok
A4	3	4	16	16	17	7 / 7	64.0 / 64.3	ok, ok
A4	3	5	32	19	20	8 / 8	302.8 / 301.9	timeout, timeout
A11	2	2	4	9	10	3 / 3	0.0 / 0.0	ok, ok
A11	2	3	8	12	13	3 / 3	0.0 / 0.0	ok, ok
A11	2	4	16	15	16	3 / 3	0.0 / 0.0	ok, ok
A11	2	5	32	18	19	4 / 4	0.1 / 0.1	ok, ok
A11	2	6	64	21	22	4 / 4	1.0 / 0.9	ok, ok
A11	2	7	128	24	25	4 / 4	3.7 / 3.6	ok, ok
A11	2	8	256	27	28	4 / 4	15.6 / 14.8	ok, ok
A11	2	9	512	30	31	4 / 4	83.0 / 78.8	ok, ok
A11	2	10	1024	33	34	4 / 4	301.0 / 301.0	timeout, timeout
A11	3	2	4	15	16	4 / 4	0.0 / 0.0	ok, ok
A11	3	3	8	20	21	4 / 4	0.1 / 0.1	ok, ok

A11	3	4	16	25	26	4 / 4	1.8 / 1.8	ok, ok
A11	3	5	32	30	31	5 / 5	300.8 / 300.8	timeout, timeout

Requested and completed

12 series, 70 cells, 140 planted systems. 116 completed, 20 stopped at the F4 budget, 4 aborted for memory (A1z m = 2 t = 7, both targets; A1g m = 3 t = 5, both targets). The limit cells are missing data, and no D enters any fit from them.

Deviations from the protocol as frozen

Amendment 1 records them all before the data: the instrument's Gebauer–Möller default, the "same cells" check done with the pilot's seed rule, the cell and target seed split, the cell-mean fit, the stop rule with a 420 s hard kill, N's chain at m = 3, and A11′s sign convention and m = 3 form. Nothing else deviated. The replication on p_2 that §5 requires for a "candidate" arm was not run in this lane.